

Delayed formation of the equatorial ridge on Iapetus from a subsatellite created in a giant impact

Andrew J. Dombard,¹ Andrew F. Cheng,² William B. McKinnon,³ and Jonathan P. Kay¹

Received 20 October 2011; revised 5 January 2012; accepted 6 January 2012; published 7 March 2012.

[1] The great equatorial ridge on Saturn's moon Iapetus is arguably the most perplexing landform in the solar system. The ridge is a mountain range up to 20 km tall and sitting on the equator of Iapetus, and explaining its creation is an unresolved challenge. Models of its formation must satisfy three critical observations: why the ridge (1) sits *exactly* on the equator, (2) is found *only* on the equator, and (3) is thus far found only on Iapetus. We argue that all previously proposed models fail to satisfy these observations, and we expand upon our previous proposal that the ridge ultimately formed from an ancient giant impact that produced a subsatellite around Iapetus. The orbit of this subsatellite would then decay, once Iapetus itself had despun due to tides raised by Saturn, until tidal forces from Iapetus tore the subsatellite apart. The resultant debris formed a transient ring around Iapetus, the material of which rained down on the surface to build the ridge. **By sequestering the material in a subsatellite with a tidally evolving orbit, formation of the ridge is delayed,** which increases the likelihood of preservation against the high-impact flux early in the solar system's history and allows the ridge to form on thick, stiff lithosphere (heat flow likely $<1 \text{ mW m}^{-2}$) required to support this massive load without apparent flexure. This mechanism thus explains the three critical observations.

Citation: Dombard, A. J., A. F. Cheng, W. B. McKinnon, and J. P. Kay (2012), Delayed formation of the equatorial ridge on Iapetus from a subsatellite created in a giant impact, *J. Geophys. Res.*, 117, E03002, doi:10.1029/2011JE004010.

1. Introduction

[2] Iapetus has proven to be one of the most peculiar bodies in the solar system. Up until recently, this third largest moon of Saturn (mean radius of 734.3 km [Thomas, 2010]) was most notable for the large semimajor axis of its orbit around Saturn (~ 59 Saturn radii) and its hemispheric albedo dichotomy, now thought to be a product of spatially variable ice mobilization on its surface [Spencer and Denk, 2010]. Beginning in 2004, however, NASA's Cassini spacecraft obtained the first high-resolution images of Iapetus, revealing a world even more peculiar than initially thought [e.g., Porco *et al.*, 2005; Denk *et al.*, 2010].

[3] Iapetus possesses a distinctly pronounced oblate spheroid shape, with an equatorial radius greater than the polar radius by $33.6 \pm 2.8 \text{ km}$ [Thomas, 2010]. This difference translates into a flattening of $\sim 4.5\%$, compared with Earth's 0.3% rotational flattening. For a body in hydrostatic equilibrium, an equatorial bulge of that size would indicate that Iapetus should be spinning once every $\sim 16.5 \text{ h}$ [Castillo-Rogez *et al.*, 2007]; this moon, however, is synchronously

locked to Saturn, rotating (and revolving) once every ~ 79 days. Models for Iapetus's flattened shape include a fossilized rotational bulge [Castillo-Rogez *et al.*, 2007; Robuchon *et al.*, 2010] and long-wavelength, axisymmetric deformation of the lithosphere [Sandwell and Schubert, 2010; Kay and Dombard, 2011].

[4] Even more bizarre than the bulge is Iapetus's peerless equatorial ridge [Porco *et al.*, 2005]. Arguably one of the most astonishing features in the solar system, it is so big that it is clearly visible in global views of the moon (Figure 1). This mountain range is up to 20 km high, 200 km wide (translating into a mass of order 0.1% the total mass of Iapetus), and sits perfectly straight, exactly on the equator. The ridge runs $>75\%$ of the circumference of the satellite [e.g., Singer and McKinnon, 2011], though not continuously, and has been modified by subsequent impacts and mass wasting (i.e., landslides [Singer *et al.*, 2009]). The cross-sectional shape is in places trapezoidal, with a flat top and sometimes a central trough, and with slopes of $\sim 15^\circ$ [Giese *et al.*, 2008]. Notably, the ridge appears to be supported by the lithosphere without an obvious flexural signal [Giese *et al.*, 2008; Dombard and Cheng, 2008] (see also below), and the rest of the surface of Iapetus is dominated by impact craters, with only a few examples of other geomorphic features that are far less impressive in scale than the ridge [Singer and McKinnon, 2011]. The ridge is heavily cratered and thus appears ancient [Denk *et al.*, 2010].

[5] Clearly, the formation of the ridge was one of the key events in the evolution of Iapetus. In this paper, we explore

¹Department of Earth and Environmental Sciences, University of Illinois at Chicago, Chicago, Illinois, USA.

²Johns Hopkins University Applied Physics Laboratory, Laurel, Maryland, USA.

³Department of Earth and Planetary Sciences and McDonnell Center for Space Sciences, Washington University, St. Louis, Missouri, USA.

[21] An additional exogenic model invokes an ancient giant impact [Levison *et al.*, 2011] (as do we). A primary objective of this work is the creation of a relatively large (0.5%–1.5% the mass of Iapetus), impact-generated “subsattellite” whose orbital evolution results in the despinning of Iapetus before (usually) reimpacting this moon. The authors, however, also speculate that remnant debris from the impact that remained within the Roche limit of Iapetus (and hence could not coalesce into a subsattellite) formed the ridge. The final stage of this proposal is similar to that of *Ip* [2006]: evolution of orbital debris to a ring over the equator and deorbiting of this material to form the ridge, although the models differ in that in the Levison scenario, the material is more tightly bound to Iapetus (i.e., within the Roche limit, or ~ 2.5 – 3 the radius of Iapetus). Their hypothesis, however, also suffers from the same shortcoming as the *Ip* [2006] proposal. As we discuss more thoroughly below, the Roche limits of most major outer satellites are well within the orbital stability zones of these moons, and all these satellites were undoubtedly pounded by large impacts toward the end of planetary accretion. Nominally, then, equatorial ridges should be common.

4. Formation via “Giant” Impact

[22] Giant impacts undoubtedly occurred in the waning stages of planetary accretion, as the largest member of an orbital zone swept up the second, third, etc., next largest members. Under proper circumstances, such large, planet-scale impacts could eject a substantial amount of material into orbit around the target body, some of which may have coalesced into a satellite. Our moon is thought to have formed this way [e.g., *Canup and Richter*, 2000]. Because of lower encounter velocities, Pluto’s moon Charon is believed to have formed via a process coined “intact capture,” where a giant impact on Pluto dissipated orbital energy, allowing the impacting body to go into orbit around Pluto [Canup, 2005, 2011]. Presumably, relatively large or “giant” impacts also occurred on the forming satellites of the outer solar system [Canup and Ward, 2006]. Here, we propose that an impact-generated subsattellite formed around Iapetus, the material in which eventually formed the equatorial ridge [cf. Dombard *et al.*, 2010].

[23] Because of the low encounter velocities of roughly coorbital bodies sharing an orbital zone with a young Iapetus about Saturn (collision speeds would be dominated by Iapetus’s escape velocity of ~ 0.5 km s^{−1}), we envision intact capture of a subsattellite, although we do not discount the possibility of coalescence from a debris disk. The resultant subsattellite does not need to be large; an icy, porous body less than ~ 100 km in radius supplies enough mass to account for the ridge. The size of the subsattellite relative to Iapetus (less than $\sim 14\%$) is smaller than the equivalent in the Earth-Moon ($\sim 27\%$) and Pluto-Charon ($\sim 52\%$) systems.

[24] The initial orbit of the subsattellite was likely fairly tight to Iapetus and eccentric. For example, the full ranges in semimajor axis and eccentricities of successful simulations by Canup [2005] were 3.7–21 Pluto radii and 0.1–0.8, respectively; results from Canup [2011] were similar. Tidal interactions with Iapetus would have initially reduced the eccentricity, circularizing the orbit, and then cause the orbital distance to evolve. Whether the orbit of Iapetus’s subsattellite expanded or decayed depended on whether the orbit was

prograde or retrograde and on the spin period of Iapetus. The orbit of a retrograde subsattellite would always decay, but a subsattellite in a prograde orbit would either retreat or decay if the period of revolution was less than the primary’s spin period (like the Earth-Moon system) or greater, respectively. Goldreich and Soter [1966] derived the time rate of change of the semimajor axis of a secondary body in orbit around a primary; under the assumption of a time invariant, dissipative elastic body, this equation can be integrated to yield

$$\bar{a} = \left(\bar{a}_0^{13/2} \pm \frac{39}{2} \frac{k_2}{Q} \frac{m}{M} \sqrt{\frac{GM}{R^3}} t \right)^{2/3}, \quad (1)$$

where $\bar{a}$ is the semimajor axis normalized to the radius of the primary (the subscripted version denotes the initial value), k_2 and Q are the second degree tidal Love number and tidal dissipation quality factor of the primary body (e.g., Iapetus), m and M are the masses of the secondary and primary bodies, G is the universal gravitational constant, R is the radius of the primary, and t is time. The orbital period is thus $2\pi\sqrt{(\bar{a}R)^3/GM}$. The k_2 of a uniform, elastic sphere is given by

$$k_2 = \frac{\frac{3}{2}}{1 + \frac{19}{2} \frac{\mu}{\rho g R}}, \quad (2)$$

where μ is the elastic shear modulus (~ 3.5 GPa for ice), ρ is density, and g is the surface gravitational acceleration; for an elastic Iapetus, k_2 is $\sim 8 \times 10^{-3}$.

[25] The plus or minus in equation (1) denotes cases of orbital retreat (+) or decay (−). Because we are considering an event occurring during planetary accretion (effectively time zero for the solar system), the spin period of Iapetus was undoubtedly much shorter than the current 79 days, likely of order 10 h [see, e.g., Castillo-Rogez *et al.*, 2007], so any subsattellite in a prograde orbit would initially retreat until Iapetus was despun, presumably by Saturn tides. (At 0.1% the mass of Iapetus, the subsattellite is too small to affect appreciably the spin of Iapetus [cf. Levison *et al.*, 2011].) Robuchon *et al.* [2010, equation 4] describes the time rate of change of the angular rotational frequency ω of a satellite, which again assuming a time invariant, dissipative elastic secondary is trivial to integrate [cf. Goldreich and Soter, 1966]:

$$\omega = \omega_0 - \frac{3}{2} \frac{k_2}{Q} \frac{GM_{GG}^2}{CM} \frac{R^3}{\alpha^6} t, \quad (3)$$

where M_{GG} is the mass of the gas giant planet and α is the semimajor axis of the satellite’s orbit around the gas giant. $\bar{C}$ is the normalized moment of inertia for the satellite (0.4 for a uniform sphere). For large icy satellites, this value ranges from ~ 0.31 (Ganymede) to ~ 0.35 (e.g., Callisto). We assume a value of 0.35.

[26] Figure 4 shows the possible configurations of our proposed Iapetus/subsattellite system, assuming a subsattellite/Iapetus mass fraction of 0.1% (comparable to the mass of the ridge), a Q for Iapetus of 100 (consistent with general

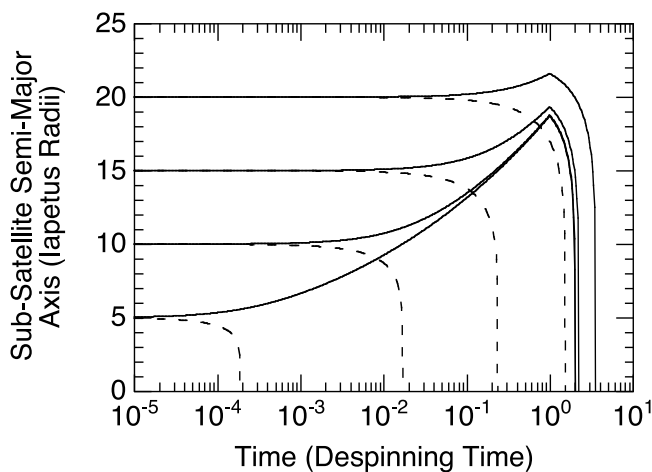


Figure 4. Tidal evolution of the semimajor axis (in Iapetus radii) of a subsatellite as a function of time for different initial values. Dashed lines show retrograde orbits and solid lines show prograde orbits. The abscissa is rendered dimensionless by dividing by the time needed for tides from Saturn to despin Iapetus to its current rotational period of 79 days. (At 0.1% the mass of Iapetus, the subsatellite is too small to appreciably affect the spin of Iapetus [cf. *Levison et al., 2011*].) For an elastic, icy Iapetus, this despinning time is ~ 7.0 Gyr. For more dissipative rheological processes (e.g., viscoelasticity), the despinning time is shortened and the orbital evolution of the subsatellite is accelerated complementarily, preserving the character of the curves. The orbits of initially close (~ 10 Iapetus radii) retrograde subsatellites will decay relatively quickly. Conversely, all prograde orbits will expand to a semimajor axis of at least 18–19 Iapetus radii until reversing direction after Iapetus despins.

expectations for solid bodies in the solar system [e.g., *Peale, 1977*]), and an initial spin period for Iapetus of 10 h. The abscissa is rendered dimensionless by dividing by ~ 7.0 Gyr, which is the time it takes for this elastic Iapetus to despin to a period of 79 days. (Despinning times for variations in the normalized moment of inertia (0.31 to 0.4) and in the initial spin period (7 to 16 h) vary from ~ 4 to ~ 10 Gyr, and the character of the curves in Figure 4 is preserved.) As discussed, retrograde orbits immediately decay, whereas prograde orbits initially retreat until Iapetus is despun, after which the orbits decay. Indeed, the total lifetimes of all possible prograde orbits exceed the age of the solar system for these nominal parameters. This is a byproduct, however, of treating Iapetus as an elastic sphere. More dissipative rheological processes, such as viscoelasticity facilitated by the thermal evolution of this satellite [e.g., *Castillo-Rogez et al., 2007, 2011; Robuchon et al., 2010*], would serve to accelerate the evolution (to timescales perhaps as short as 0.9 Gyr [*Castillo-Rogez et al., 2011*]). These effects would violate the simple integrations used to produce equations (1) and (3) by introducing a time dependency to the k_2 and Q of Iapetus; however, both equations (1) and (3) and their root differentials depend on these factors in the same way. Thus if the despinning of Iapetus is accelerated, the orbital evolution of the subsatellite changes in lock step with the despinning. In essence, the abscissa

of Figure 4 is placed on a nonlinear sliding bar. Thus, the salient points of Figure 4 are that (1) initially close (less than $\sim 10 R_I$ (Iapetus radii)) retrograde orbits would decay rather quickly and (2) all prograde orbits should expand to a semimajor axis greater than ~ 18 – $19 R_I$ before reversing direction. As we will discuss in section 5, this distance is still within Iapetus's gravitational zone of dominance.

[27] The subsatellite, whether initially prograde or retrograde, would have ultimately encountered Iapetus's Roche limit of at ~ 2.5 – $3 R_I$. We anticipate that the subsatellite would have accreted as a rubble pile, and even if not, have been pervasively fractured during its impact formation/capture. Thus inward of ~ 2.5 – $3 R_I$, tidal forces should ultimately overwhelm the subsatellite, tearing it apart. The resultant debris would then collisionally evolve, which would (1) comminute the material, in principle; (2) dissipate any remaining inclination in the subsatellite's orbit, placing the debris into a ring over the equator; and (3) dissipate orbital energy. For a ring mass of 0.1% that of Iapetus, the surface mass density and viscosity would be far higher than that of Saturn's main rings, which would lead to very rapid settling of debris into Iapetus's equatorial plane and viscous spreading within that plane [e.g., *de Pater and Lissauer, 2010, section 11.2*]. The endgame of our scenario is very similar to that proposed by *Ip [2006]*, with a debris ring mostly deorbiting and impacting the surface of Iapetus at very shallow angles and subsonic speeds of a few hundred meters per second. Over time, the repeated impacts added material to the surface in a narrow zone, building the ridge.

5. Discussion

[28] Our hypothesis for the formation of the equatorial ridge explains the three critical observations. The first two (why on and only on the equator) are explained via the hypothesis's penultimate step as a debris ring. The last observation (why only Iapetus) is a natural consequence of the satellite's unique orbital position far from its gas giant parent. Figure 5 shows the zones of prograde orbital stability for all the major satellites of the outer solar system (i.e., those large enough to be roughly spherical), calculated by determining the semimajor axis at which is reached a minimum nondimensional Jacobi constant [see *Hamilton and Burns, 1991, equation 5*]. Above a critical value (~ 9.4), orbital trajectories are circular and stable, and at somewhat smaller values (between 9 and ~ 9.4), trajectories can become chaotic. For values < 9 , the Hill curves no longer fully enclose the satellite, meaning any subsatellite may (though not necessarily will) be lost to an orbit around the gas giant planet. The Jacobi constant also depends on orbit inclination; thus the ranges in Figure 5 are found by considering the most restrictive (critical Jacobi constant of 9 and inclination of 90°) and forgiving (9.4 and 0°) cases. Also plotted are the Roche limit for each satellite and the range in Charon's semimajor axis in the *Canup [2005]* simulations (in satellite radii), which we take to be representative of intact capture. Although all the satellites in Figure 5 likely suffered giant impacts, Iapetus clearly has more stable phase space than any other satellite; many satellites only have stable orbital zones within their Roche limits. At best they could only retain a postimpact debris disk, and it is uncertain that the amount of retained debris could yield a ridge (we consider it unlikely, or at least